# Sensory Characterization of an Active-Passive Exoskeleton For Human Gait

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***Abstract*—Exoskeletons are wearable devices designed to enhance the physical capabilities of the user. Currently, research on these devices focuses on providing assistance to individuals with disabilities to improve their quality of life. In this context, lower-limb mobility represents a critical issue due to the large population affected. This paper proposes the development of a lower-limb exoskeleton sensory system designed to feed a mechanical system for human gait. The sensing framework measures joint position, torque, plantar pressure, and trunk inclination to detect movement intention (passive assistance) and propose exoskeleton assistance (active assistance). The system incorporates non-invasive mechanical and inertial sensors integrated into an electronic architecture based on a real-time microcontroller, a high-level processing unit, and $I^2C$ & CAN bus communication. Additionally, electronic safety mechanisms are integrated to ensure reliable operation during rehabilitation tasks.**

***Index Terms*—Lower-limb exoskeleton, sensory characterization, gait analysis, sensor fusion, passive assistance mode, wearable sensors.**

## I. Introduction

Assistive robotics has emerged as a key field focused on the development and deployment of exoskeleton systems, which demand precise kinematic and dynamic coupling with human physiology (Zhang et al., 2025). The design and implementation of exoskeletons encompass diverse applications, ranging from virtual reality and adaptive training to medical conditioning and the rehabilitation of individuals with motor disabilities. To ensure functionality, these wearable devices must integrate biomechanical sensor networks on the user's body to capture physiological signals and motion intent. In exoskeleton design, three main categories are identified based on the mode of assistance, which also dictate the complexity of the control architecture: passive, active, and active-passive (hybrid) systems. Passive systems rely on elastic elements designed to store and release biomechanical potential energy during human gait. These devices are applied in exoskeletons to mitigate muscle stress and fatigue by reducing the physical load and metabolic cost demanded by various tasks. Structurally, these mechanisms typically incorporate springs, dampers, elastic bands, counterweights, or pneumatic actuators for gravitational compensation. Active exoskeletons, also referred to as active orthoses (Faqiri et al., 2023), provide dynamic, time-independent assistive torques, in contrast to passive exoskeletons, whose torques directly depend on the instantaneous joint angle and the elastic properties of the mechanism. Active-Passive Exoskeletons, also referred to as semi-active, semi-passive, or quasi-passive systems, combine both power transmission approaches to mitigate the limitations of purely active or passive configurations (Al-Hayali et al., 2021). On one hand, passive mechanisms reduce the system's size and weight while lowering the required active torque and improving overall portability. On the other hand, the active actuators supply supplementary assistive torque and can harvest energy during negative joint work, thereby extending the operational runtime by more than $20\%$ (Ren et al., 2021).

To address these challenges prior to closed-loop actuator deployment, the main objective of this work is to design, characterize, and experimentally validate a deterministic, hard real-time multimodal sensory and state-estimation architecture for the SARA active-assistive lower-limb exoskeleton. The primary contributions of this paper are: 1. Deterministic Hard Real-Time Acquisition Architecture: The implementation of a non-blocking state-machine multiplexing scheme and Direct Memory Access (DMA) binary framing on a dual-core STM32H755ZI microcontroller, achieving a certified Worst-Case Execution Time (WCET) of $0.65$ ms within a $100$ Hz ($10$ ms) control loop and a $0.00\%$ packet-loss rate. 2. Rigorous Metrological Sensor Characterization: The empirical quantification of triaxial IMU noise parameters via Allan Variance analysis alongside a stochastic ($3\sigma$) baseline thresholding and quasi-static gravimetric calibration ($K = 0.002703$ N/LSB) for a 12-sensel plantar FSR matrix. 3. Real-Time Sensor Fusion and Sagittal Kinetic Estimation: The integration of an Extended Kalman Filter (EKF) coupled with FSR-driven Zero-Velocity Updates (ZUPT) and analytical Center of Pressure ($CoP_y$) tracking to compute drift-free sagittal joint kinematics and quasi-static ankle moments during continuous human gait.

## II. Data Acquisition Technologies and Motion-Intention Sensors

Capturing motion intent and the dynamic state of the body is fundamental for human-robot synchronization. Historically, surface electromyography (sEMG) has dominated this field due to its ability to detect muscle activation before mechanical movement occurs (Neťuková et al., 2022). However,

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relying exclusively on sEMG presents significant limitations, as these signals are susceptible to skin impedance variation, sweat, muscle fatigue, electromagnetic interference, and electrode displacement; this demands complex calibration routines and restricts their reliability outside laboratory environments (Coser et al., 2025; Costa et al., 2026).

In contrast, wearable architectures have evolved to rely predominantly on the multimodal integration of inertial measurement units (IMUs) and mechanical sensors—such as encoders and force-sensitive resistors (FSRs)—which provide highly reactive and robust kinematic and kinetic measurements (Manna et al., 2023; Yao et al., 2024). The literature categorically establishes that combining IMU and FSR sensors captures a significantly more comprehensive and accurate data spectrum than deploying either sensor in isolation (Manna et al., 2023; Yao et al., 2024).

Despite the benefits of multimodal fusion, Wang et al. (2023) highlights the critical challenge of achieving strict synchronization across multimodal data streams. In their study, latencies of up to 200 ms were recorded among sensory readings due to the non-deterministic nature of wireless communication protocols such as TCP/UDP, Wi-Fi, and Bluetooth. Consequently, offline synchronization techniques were required to compensate for errors when computing joint torques (Wang et al., 2023). Therefore, implementing a hard real-time (HRT) system ensures the determinism required to simultaneously acquire inertial data, plantar pressure, and joint proprioception, guaranteeing the temporal coherence indispensable for inverse dynamics computation.

## III. Proposed Sensory System Architecture

The SARA exoskeleton is a mechanical device designed to assist lower-limb motion in individuals with gait impairments. Its structural design is conceived to couple externally to the user's body, ensuring alignment with the primary joint axes of the legs to facilitate controlled movement during locomotion and biomechanical rehabilitation protocols.

Figure 1 illustrates the overall architecture of the exoskeleton integrated with a human anthropometric model, highlighting the spatial distribution of the mechanical modules and their kinematic alignment with the user's principal anatomical joints.

### A. Electronic Architecture and Signal Processing

The electronic system of the SARA lower-limb exoskeleton is designed to acquire the user's biomechanical data, process it in real time, and deliver targeted assistance through electrical actuators. To achieve this, the platform integrates a sensor network comprising inertial measurement units, as well as position, torque, and plantar force sensors, coupled with high-performance processing units and communication modules that ensure deterministic, safe, and reliable operation.

### B. System Sensors

The exoskeleton's sensors measure the physical variables required to monitor the user's state and enable assisted motion control. These sensors are non-invasive and were selected based on criteria of reliability, portability, safety, and user comfort.

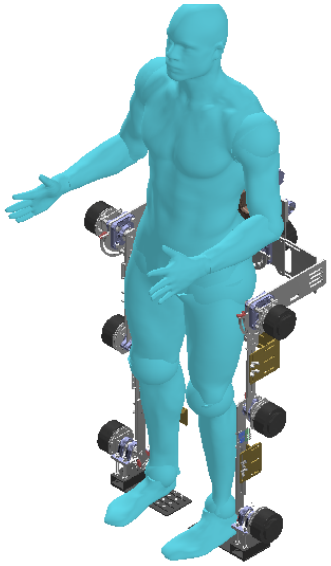


Fig. 1. SARA Exoeskeleton

TABLE I
Mechatronic and Sensory Instrumentation of the SARA Exoskeleton

| Sensor Type | Model / Tech. | Main Function | Location | Qty. |
|---|---|---|---|---|
| Absolute Encoder | Integrated (T-Motor AK80-64) | Joint kinematics acquisition, gait phase, and position control | Hip, knee, and ankle | 8 |
| Torque Sensor | Integrated (T-Motor AK80-64) | Motor torque feedback for adaptive assistive control | Actuated joints | 8 |
| Plantar Force | Piezoresistive (FSR-402) | Ground reaction force (GRF) and contact phase detection | Plantar insole | 24 |
| Limit Switch | Optical (OP5315W2) | Range-of-motion enforcement and active mechanical bumpers | Segmental joints | 16 |
| Inertial (IMU) | 6-DoF MEMS (MPU-6050) | Segmental orientation, angular velocity, and motion intent | Pelvis and thighs | 3 |
| Pushbutton | Normally-Closed (NC Button) | Emergency shutdown (Safe Torque Off) and logic power | Dorsal frame | 2 |

Figure 2 illustrates the schematic kinematic topology of these sensory modules and actuators across the lower-limb segments, while Table I summarizes the specific hardware models, quantities, and functional roles. At the physical layer, the two dorsal emergency pushbuttons are wired to execute an immediate hardware-level power cutoff (Safe Torque Off) upon activation. Simultaneously, the 16 optical limit switches (OP5315W2) act as active mechanical-range bumpers posi-

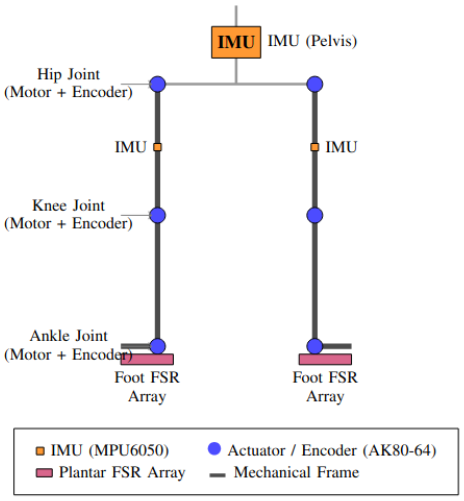



Fig. 2. Schematic representation of sensor placement and joint actuator distribution on the lower-limb exoskeleton.

tioned at the anatomical extremes of hip, knee, and ankle flexion/extension that inhibit actuator torque generation when triggered.

### C. Control and Processing Units

The exoskeleton's electronic system implements a hierarchical control architecture comprising a real-time microcontroller and a high-level processing unit.

The STM32H755ZI microcontroller manages sensor data acquisition and low-level, deterministic real-time control of the actuators. This device features a dual-core architecture, where the ARM Cortex-M7 core executes the motor control loops, while the ARM Cortex-M4 core is dedicated to acquiring and preprocessing sensory signals.

The high-level processing unit consists of a ZED Box computer, which executes sensor fusion algorithms and gait-phase estimation. Inter-processor communication between the microcontroller and the ZED Box is established via a USB Virtual COM Port (VCP).

## IV. Data Adquisition

### A. Force and Pressure Sensors

To estimate ground reaction forces (GRF) during locomotion, FSR-402 pressure sensors are embedded within the insole of each foot. Twelve sensors per foot are strategically distributed to detect the stance and swing phases of the gait cycle, as depicted in Fig. 3.

To eliminate spurious activations arising from passive mechanical pre-loading and shoe deformation, an empirical noise characterization was conducted under no-load conditions, yielding the baseline mean $\mu_k$ and standard deviation $\sigma_k$ per sensel. Contact detection enforces a $3\sigma$ confidence bound:

$$\text{Thresh}_k = \mu_k + 3\sigma_k + \Delta_{\text{guard}} \tag{1}$$

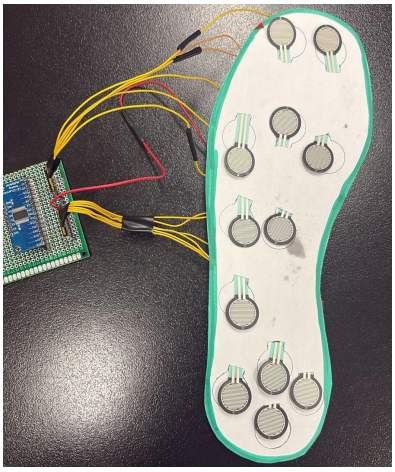


Fig. 3. Sensor distribution on an insole.

The stochastic baseline evaluation shown in Fig. 4(a) quantifies the intrinsic noise floor across the 12 unloaded plantar sensels ($N = 824$ frames, 14.97 s). The raw baseline means remained tightly bounded within $\mu_k \in [0.08,\ 0.93]$ LSB ($\sigma_k \leq 1.65$ LSB). Enforcing the adaptive activation threshold in (1) with $\Delta_{\text{guard}} = 15.0$ LSB establishes channel cutoffs of $\text{Thresh}_k \in [16.13,\ 20.87]$ LSB, surpassing the maximum observed unloaded transient (15.0 LSB on $S_2$) and achieving a 0.00% false-positive stance rate.

Figure 4(b) illustrates the dynamic step response under an abrupt heel impact held for 8.07 s ($t \in [3.44,\ 11.51]$ s). Auditing the hardware cycle counter (`DWT->CYCCNT` at 480 MHz) confirms a 10%–90% elastomeric rise time of $t_r = 14.60$ ms on the lateral heel ($S_{11}$) and 29.33 ms on the posterior calcaneus ($S_{12}$), settling into a viscoelastic plateau of $25{,}542.3 \pm 1{,}991.9$ LSB. Concurrently, the unloaded midfoot and forefoot channels ($S_1$–$S_8$) exhibited a mean cross-coupling leakage of only 1.75 LSB, demonstrating a spatial mechanical isolation of 82.58 dB between anatomical subareas.

The net vertical Ground Reaction Force ($F_z$) is derived via a quasi-static gravimetric gain factor $K_{\text{force}}$ which calibration curve is presented in Fig. 4(c) during full monopodal body-weight support ($m = 72.0$ kg, target reference $F_{\text{ref}} = 706.08$ N).

$$F_z(t) = K_{\text{force}} \sum_{k=0}^{11} \max\left(0,\ \text{ADC}_k(t) - \mu_k\right) \tag{2}$$

Whenever $F_z(t) \geq F_{\text{min,stance}}$ (35.0 N) or any structural sensel satisfies $\text{ADC}_k(t) \geq \text{Thresh}_k$, the boolean stance flag $\gamma_{\text{stance}}(t) \in \{0, 1\}$ is asserted. Consequently, the longitudinal Center of Pressure ($CoP_y$) and sagittal quasi-static ankle joint moment ($\tau_{\text{ankle}}$) are computed deterministically as:

$$\begin{aligned} CoP_y(t) &= \frac{\sum_{k=0}^{11} y_k\, f_k(t)}{\sum_{k=0}^{11} f_k(t)} \\ \tau_{\text{ankle}}(t) &= \left(CoP_y(t) - y_{\text{ankle}}\right) F_z(t) \end{aligned} \tag{3}$$

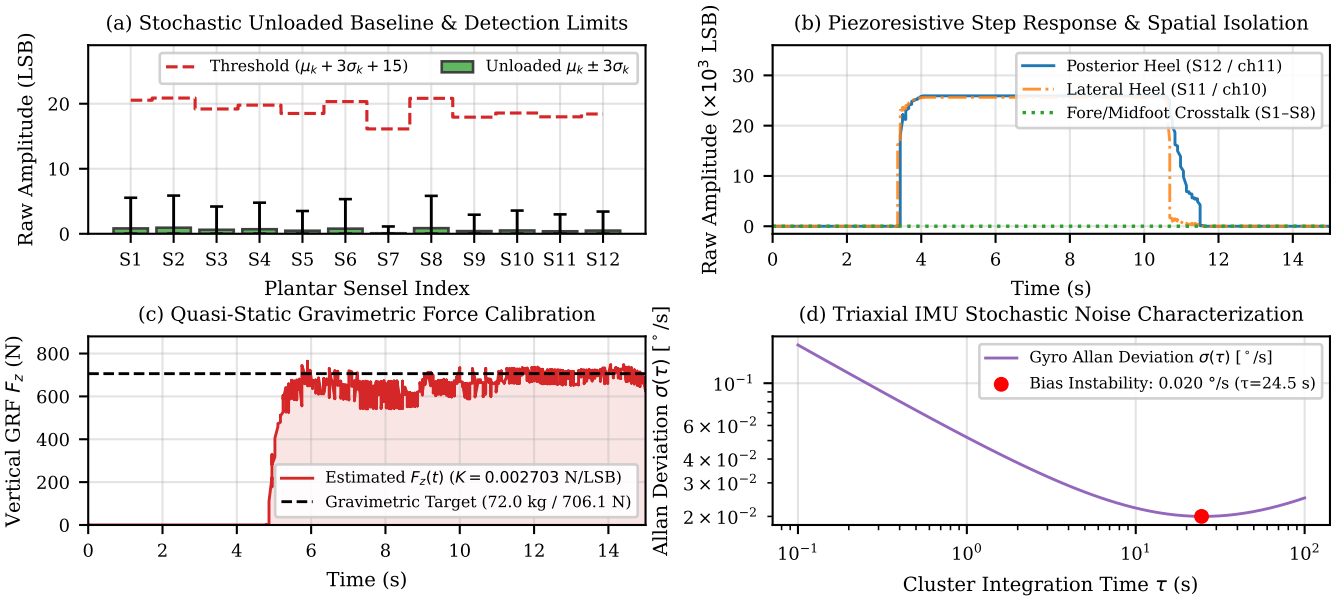



Fig. 4. Characterization and calibration pipeline of the multimodal sensory architecture: (a) Unloaded stochastic baseline means ($\mu_k$) and three-sigma detection boundaries ($\mu_k \pm 3\sigma_k$) across the 12-channel FSR matrix with adaptive activation thresholds ($\text{Thresh}_k$), (b) Transient step response under abrupt heel impact illustrating elastomeric rise time and high spatial isolation (82.58 dB) against midfoot and forefoot channels, (c) Quasi-static gravimetric calibration during monopodal full body-weight plateau loading validating the linear force constant ($K_{\text{force}} = 0.002703$ N/LSB) against the 706.1 N gravitational reference, and (d) Allan deviation $\sigma(\tau)$ of the triaxial rate gyroscopes characterizing white noise Angle Random Walk and bias instability floor.

where $f_k(t) = K_{\text{force}} \max(0, \text{ADC}_k(t) - \mu_k)$ for active sensels.

### B. Inertial Sensors (IMUs)

Data acquisition is performed using an STM32 development board, which manages sensor sampling, initial signal preprocessing, and data transmission to the high-level processing unit. Each MPU6050 sensor integrates a triaxial accelerometer and a triaxial gyroscope, enabling the measurement of both linear acceleration and angular velocity for its corresponding body segment. The system incorporates three sensors distributed as follows:

- **Sensor 0:** Positioned on the lower back (pelvis/hip region).
- **Sensor 1:** Positioned on the right thigh.
- **Sensor 2:** Positioned on the left thigh or right shank.

*1) Sensor Data Acquisition Routine:* Each sampling frame contains 42 variables: 18 kinematic variables from the IMUs and 24 kinetic readings from the plantar FSR arrays (12 per foot). During each $I^2C$ transaction, 14 consecutive bytes are retrieved, corresponding to: Triaxial acceleration ($X, Y, Z$ axes), Triaxial angular velocity ($X, Y, Z$ axes) and Internal sensor temperature.

For the accelerometer, raw values are normalized by 16384 LSB/$g$ to yield acceleration in gravity units ($g$). For the gyroscope, a scale factor of 131 LSB/(°/s) is applied to obtain the angular velocity in degrees per second (°/s). These values are reported through an implemented deterministic binary architecture that works with a non-blocking asynchronous state machine and $I^2C$ sprees bounded by 2ms timeouts, which are gathered into 117 Byte packages and transmitted via DMA.

*2) Allan Variance and Triaxial Offset Calibration:* Overlapping Allan Deviation (OAVAR) analysis at 100 Hz across the three segments—summarized in Fig. 4(d)—confirmed a white-noise Angle Random Walk (ARW) of $0.02°$–$0.05°/\sqrt{\text{s}}$ and a gyroscope bias instability floor of $B_\omega \in [0.002, 0.020]°/\text{s}$ within $\tau \in [24.5, 120]$ s, alongside an accelerometer Velocity Random Walk (VRW) of $3 \times 10^{-4}\, g/\sqrt{\text{Hz}}$ and bias floor of $B_a \in [25, 65]\, \mu g$. Static thermal-equilibrium calibration ($T_{\text{amb}} \in [26.6, 27.7]°\text{C}$) quantified the triaxial gyroscope offsets depicted in Fig. 5(d), revealing substantial intrinsic biases—most notably $b_{\omega,y} = -7.43 \pm 0.18°/\text{s}$ on the thigh and $b_{\omega,x} = -5.21 \pm 0.16°/\text{s}$ on the shank— alongside low accelerometer noise floors ($\sigma_a \in [0.0030, 0.0153]\, g$).

*3) Real-Time Sampling Loop:* To validate hard real-time determinism prior to closed-loop actuator deployment, the primary control and sensory acquisition thread is executed at an isochronous rate of 100.0 Hz ($T_{\text{period}} = 10.00$ ms) on the dedicated Arm Cortex-M7 core running at 480 MHz. Execution latencies and deterministic phase bounds are audited at single-cycle resolution using the hardware Data Watchpoint and Trace cycle counter register (`DWT->CYCCNT`).

## V. Data Processing

Experimental trials were conducted on a healthy male volunteer ($N = 1$, mass: 72.0 kg, height: 1.72 m, foot length: $L_{\text{foot}} = 0.2615$ m, ankle axis: $y_{\text{ankle}} = 0.0550$ m) across four standardized biomechanical conditions totaling 11,374 synchronized frames:

- **Condition 1 (Orthostatic Stance, 24.7 s):** Quiet upright standing to audit baseline angle stability, static Center of Pressure ($CoP_y$), and gravimetric load tracking.

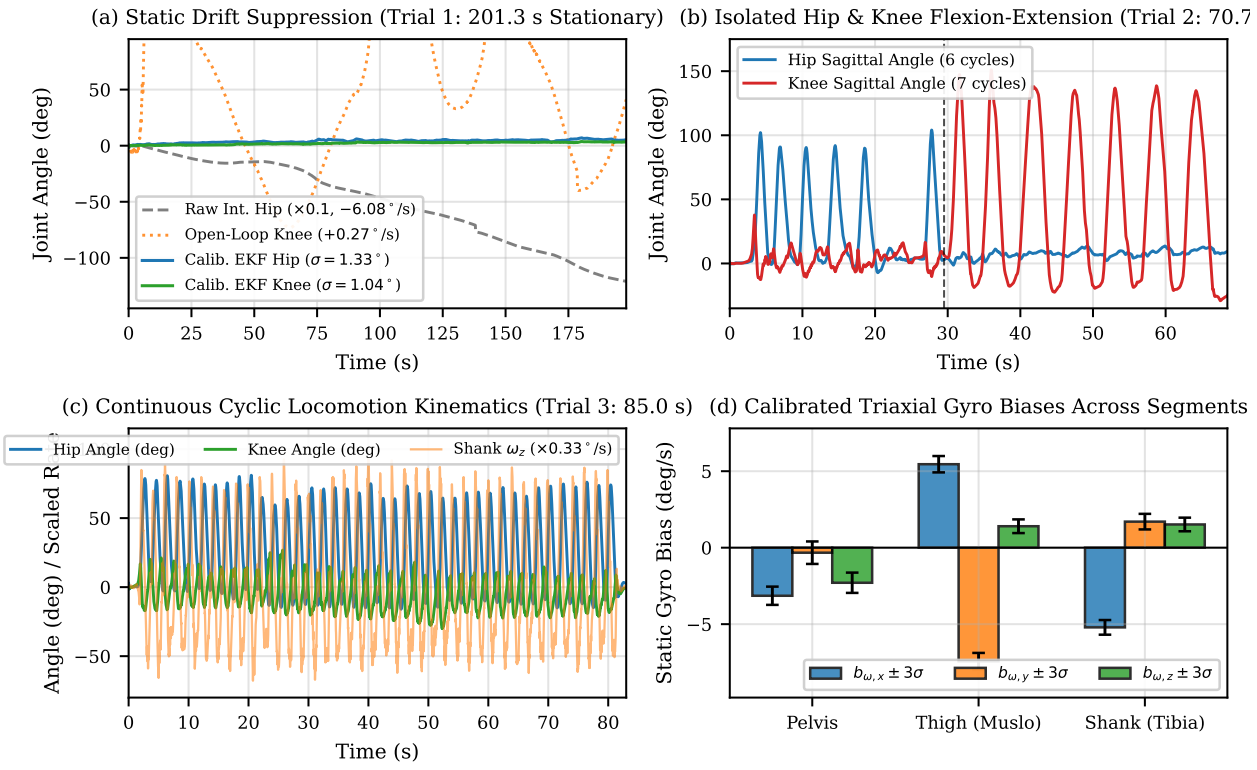



Fig. 5. Experimental validation of the 3-IMU kinematic estimation pipeline at 100.0 Hz ($N = 53{,}880$ total frames): (a) Stationary 201.3-s benchmark (Trial 1) demonstrating complete eradication of open-loop dead-reckoning drift ($-6.08^\circ$/s on hip) down to $\sigma_{\text{hip}} = 1.33^\circ$ and $\sigma_{\text{knee}} = 1.04^\circ$, (b) Isolated sequential multi-joint flexion-extension (Trial 2, 70.7 s) showing 6 hip cycles ($104.1^\circ$ peak) decoupled from 7 deep knee flexion cycles ($151.1^\circ$ peak), (c) High-speed continuous cyclic stepping (Trial 3, 85.0 s, 26 cycles) tracking coupled hip/knee trajectories under shank angular rates up to $280.2^\circ$/s, and (d) Empirically calibrated triaxial gyroscope offsets ($\mathbf{b}_\omega \pm 3\sigma$) across pelvis, thigh, and shank segments.

- **Condition 2 (Unilateral Swing Suspension, 21.8 s):** Transition from stance to aerial limb suspension ($t \in [5.6,\ 16.5]$ s) with active knee flexion-extension in the air, followed by ground re-contact, to verify immunity against false-positive stance chattering.
- **Condition 3 (Dynamic Sagittal Roll-Over, 38.0 s):** Cyclic heel-strike, mid-stance, and forefoot push-off progressions to validate longitudinal $CoP_y$ migration and sagittal ankle moment ($\tau_{\text{ankle}}$) polarity inversion.
- **Condition 4 (Continuous Overground Gait and Terminal Stop, 132.6 s):** Prolonged comfortable walking ($\approx 125$ s) concluded by an abrupt stationary standing stop ($t \geq 126.5$ s) to assess EKF-ZUPT drift bounding.

The quantitative results across all four conditions are synthesized in Table II, while representative time-series profiles are depicted in Fig. 6. Across all 11,374 frames, hardware cycle profiling certified a mean execution time of $0.6480 \pm 0.0002$ ms and an absolute peak of WCET $= 0.6496$ ms, confirming strict hard real-time determinism (93.5% CPU slack) and 0.000% packet loss.

## VI. Conclusions and Future Work

This paper presented the metrological characterization, hard real-time embedded architecture, and multimodal kinematic-kinetic state-estimation framework for the SARA active-assistive lower-limb exoskeleton. By integrating an asynchronous non-blocking FSR state machine, bounded $I^2C$ inertial burst reads, and binary DMA framing on an STM32H755ZI microcontroller, the platform achieved a certified WCET of 0.65 ms (93.5% computational margin) and 0.000% packet loss. Coupling the calibrated 12-sensel plantar matrix (82.58 dB spatial isolation, 4.63% gravimetric RMSE) with a bias-compensated sagittal EKF enabled drift-bounded joint tracking, longitudinal $CoP_y$ migration analysis, and quasi-static ankle torque estimation during continuous human locomotion.

Despite these validation milestones, three methodological shortcomings of the present study must be noted: *(i)* experimental trials were limited to a single healthy participant ($N = 1$) in a proof-of-concept setting, requiring future clinical validation across cohorts with gait pathologies; *(ii)* sagittal ankle torque was derived via a quasi-static planar projection without full 3D inertial link-segment accelerations or optical motion-capture ground truth; and *(iii)* the USB-VCP serial link capped host logging throughput at $\approx 50.3$–$53.8$ Hz, while prolonged walking in Condition 4 exposed $I^2C_4$ harness strain after 97.1 s. Future work will transition the telemetry link to high-speed CAN-FD, reinforce physical sensor shielding,

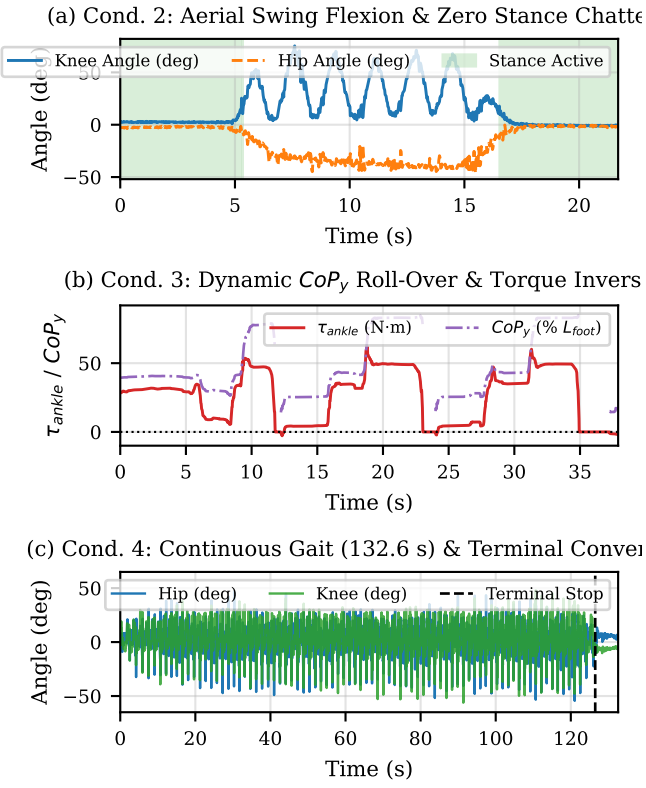



Fig. 6. Multimodal human-in-the-loop validation across dynamic conditions: (a) Condition 2 (aerial swing knee flexion up to 75.3° exhibiting 0.00% false-positive stance chattering during suspension), (b) Condition 3 (dynamic sagittal roll-over illustrating $CoP_y$ forward migration and $\tau_{\text{ankle}}$ polarity inversion from $-2.80\,\text{N}\cdot\text{m}$ dorsiflexor braking to $+62.23\,\text{N}\cdot\text{m}$ plantarflexor propulsion), and (c) Condition 4 (132.6 s continuous walking trial followed by a terminal stationary stop confirming drift-bounded convergence).

and deploy deep-learning gait-phase classifiers (LSTM) on the ZED Box to close the active-assistive torque control loop.

## Acknowledgment

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TABLE II
Quantitative Multimodal Performance Across Experimental Conditions

| **Metrological Metric** | **Cond. 1** | **Cond. 2** | **Cond. 3** | **Cond. 4** |
|---|---|---|---|---|
| Biomechanical Regime | Static | Swing | Roll-Over | Gait+Stop |
| Trial Duration ($s$) | 24.66 | 21.78 | 37.97 | 132.63 |
| Recorded Frames ($N$) | 1,240 | 1,095 | 1,908 | 7,131 |
| Mean / Max WCET ($ms$) | 0.648/0.648 | 0.648/0.649 | 0.648/0.648 | 0.648/0.650 |
| Packet Loss Rate (%) | 0.000 | 0.000 | 0.000 | 0.000 |
| Active Stance Ratio (%) | 100.0 | 0.00* | 90.2 | 67.1† |
| Peak Vertical GRF $F_z$ ($N$) | 607.4 | 0.0* | 660.9 | 733.1 |
| $CoP_y$ Span (% $L_{\text{foot}}$) | 36.4–39.9 | N/A* | 14.2–84.8 | 13.6–92.6 |
| $\tau_{\text{ankle}}$ Range ($N\cdot m$) | +21.9–30.0 | 0.0* | −2.8– + 62.2 | −3.1– + 64.2 |
| Hip Angle Span (°) | $0.1 \pm 0.7$ | −46.2–0.0 | +7.9– + 34.3 | −54.5– + 46.8 |
| Knee Angle Span (°) | $-0.4 \pm 0.3$ | −1.6– + 75.3 | −22.9– + 12.7 | −56.2– + 48.1 |

*Evaluated during the pure aerial suspension window ($t \in [6.0, 16.0]$ s). †Active bus window (0–97.1 s).


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